Data Structures and Algorithms
Practice Questions

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Problem Statement:

Given a string and the task is to count vowels, consonant, digits and special character in string. Special character also contains the white space.

Input : str = "peers got geers121"
Output : Vowels: 5
Consonant: 8
Digit: 3
Special Character: 2

Input : str = "_A1 B@ d _adc"
Output : Vowels: 2
Consonant: 4
Digit: 1
Special Character: 5

T.C.	O(n)
S.C.	O(1)

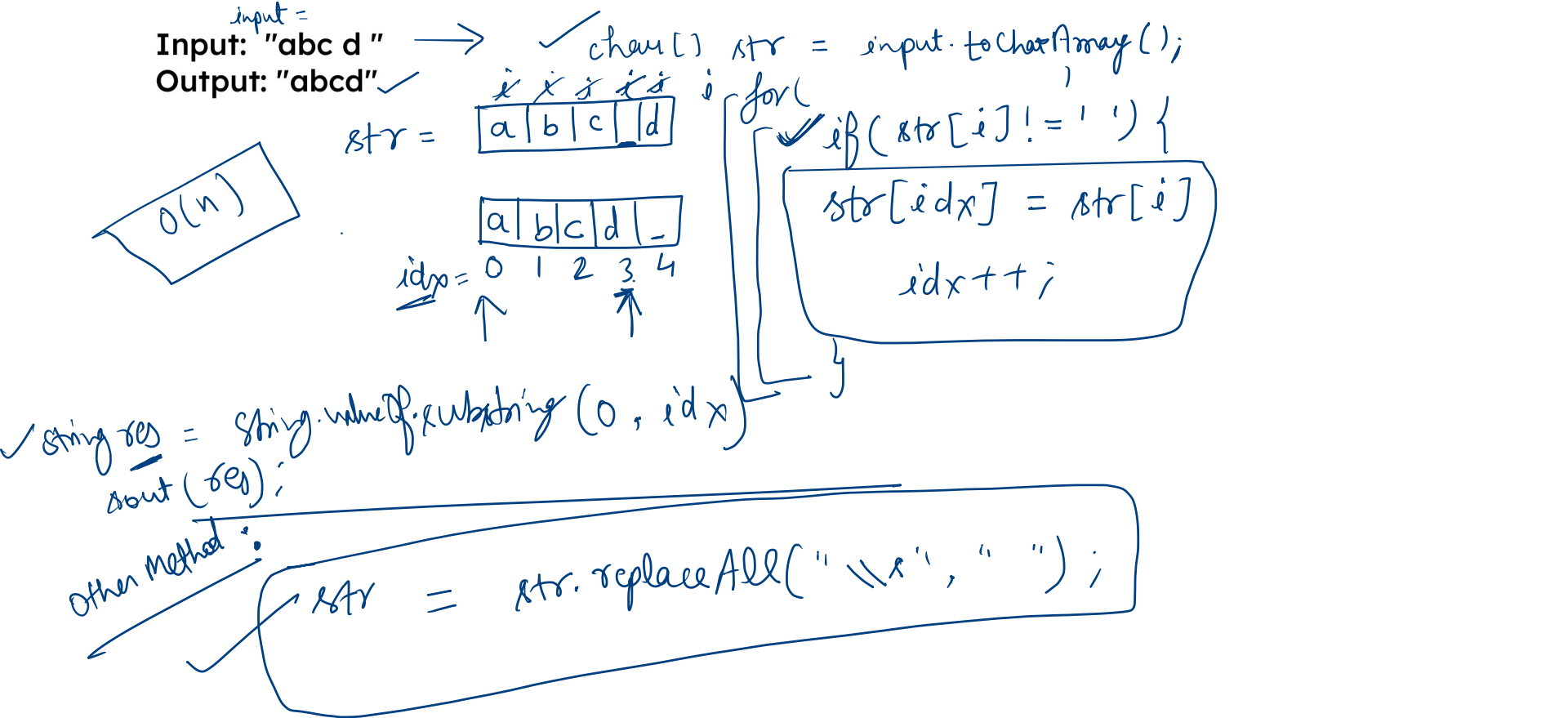
```
Vowels = A E I O U
         a e i o u

int v=0, c=0, d=0, sc=0
for(i=0; i<str.length(); i++) {
    char ch = str.charAt(i);
    if((ch>'a' && ch<='z') ||
       (ch>'A' && ch<='Z')) {
        ch = Character.toLowerCase(ch);
        if(ch=='a' || ch=='e' ||
           ch=='i' || ch=='o' ||
           ch=='u') {
            v++;
        }
        else {
            consonant++;
        }
    }
    else if(ch>='0' && ch<='9') {
        digit++;
    }
    else {
        sc++;
    }
}
```

Problem Statement:

Given a string, remove all spaces from the string and return it.

Input: "abc d "
Output: "abcd"



Problem Statement:

Given a string s, the task is to identify all characters that appear more than once and print each as a list containing the character and its count.

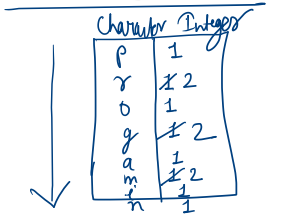
Input: s = "programming"
Output: ['r', 2], ['g', 2], ['m', 2]

Input: s = "mississippi"
Output: ['i', 4], ['s', 4], ['p', 2]

T.C. O(n)

S.C. O(n)

p r o g r a m m i n g




```
HashMap<Character, Integer> map = new HashMap<>();
for(char c : s.toCharArray()) {
    map.put(c, map.getOrDefault(c,0)+1);
}

for(Map.Entry<Character, Integer> entry : map.entrySet()){
    if(entry.getValue() > 1) {
        sout("[" + entry.getKey() + ","
            + entry.getValue() + "],");
    }
}
```